

Ablation summary: which “knob” decides adversarial robustness?

Factor we varied	Held fixed	Result	What we learned
Add vs. remove (FoSR vs. ProxyDelete)	over-squashing goal	both fail to defend	direction is not what matters
Edit criterion (spectral / similarity / low-rank)	that we edit edges at all	only attack-aware editing defends (Jaccard, SVD)	the criterion is the key
Edge budget (253 vs. 643 edges)	the method	spectral removal still fails	it is not a budget artifact
Dataset (Cora vs. Citeseer)	the method	same ranking of methods	the finding generalizes
Attacker (non-adaptive vs. adaptive)	the defense	evades the filter but turns weak	potency & detectability are coupled
Over-squashing measure (clean vs. poisoned graph)	the model	decay rate unchanged	over-squashing & fragility are decoupled